Linearity in Federbush's Genius Conjecture

MathDB 345451

Principia Math In Progress

September 18, 2026

Abstract

A formal change of variable absorbs the exponential factor independent of y in Federbush's coefficient identity. The resulting functions are coefficients of a composition with the original series and are therefore linear in its coefficients. An explicit Lagrange coefficient formula is obtained at the same time.

Task 1. Full solution

The source's identity

Fix a positive integer p and set

$$U(x)=\sum_{k=1}^{p} u_k x^k, \quad u_1=1, \qquad D(x)=\sum_{k=1}^{p} d_k x^k, \qquad A(x)=e^{D(x)}=\sum_{k\ge 0} a_k x^k.$$

Assume $a_p = 0$, as in equation (1) of [1]. Theorem 1.1 of that paper defines $F_1 = 1, F_2, \dots, F_p$ uniquely by

$$[x^p]A(x)e^{yU(x)} = [t^p]\exp\left(y\sum_{i=1}^{p} F_i t^i\right). \tag{1}$$

Conjecture 1 asks whether each F_i is linear in the u_k. Here $u_1 = 1$ is included among the coefficients: equivalently, the dependence on $u_2, \dots, u_p$ is affine, with no products of those variables. We prove the stronger explicit formula below.

Theorem 1. *Define the polynomial*

$$W_p(x)=\sum_{k=0}^{p-1}\frac{p}{p-k}a_k x^k. \tag{2}$$

Then for every $1 \le i \le p$,

$$\boxed{F_i=\frac{1}{i}[x^{i-1}]U'(x)W_p(x)^{i/p}=\sum_{k=1}^{i}\frac{k}{i}u_k[x^{i-k}]W_p(x)^{i/p}.} \tag{3}$$

In particular, Conjecture 1 of [1] holds for every p.

All fractional powers are formal binomial series with constant term 1. The coefficients a_k depend only on the d_k, so (3) displays the asserted linear dependence directly.

Absorbing the weight by a change of variable

Work over the characteristic-zero coefficient ring generated by the u_k, d_k, subject to $a_p = 0$. Since $a_0 = 1$, define

$$\phi(x) = xW_p(x)^{-1/p} = x + O(x^2).$$

It has a unique compositional inverse $\psi(t) = t + O(t^2)$. Differentiating $\phi(x)^{-p} = x^{-p}W_p(x)$ gives the exact identity

$$\begin{aligned}\frac{\phi'(x)}{\phi(x)^{p+1}} &= x^{-p-1}\left(W_p(x) - \frac{x}{p}W_p'(x)\right)\\ &= x^{-p-1}\sum_{k=0}^{p-1} a_k x^k. \end{aligned} \tag{4}$$

For a formal Laurent series, Res_x denotes its coefficient of x^{-1}. Let $H(x)$ be any formal power series. Because $a_p = 0$, terms of $A(x)$ beyond those in (4) have degree at least $p+1$ and cannot contribute to the residue below. The usual formal change-of-variable rule consequently gives

$$\begin{aligned}[x^p]A(x)H(x) &= \mathrm{Res}_x\, H(x)\frac{\phi'(x)}{\phi(x)^{p+1}}\\ &= \mathrm{Res}_t\, \frac{H(\psi(t))}{t^{p+1}} = [t^p]H(\psi(t)). \end{aligned} \tag{5}$$

This rule can be verified on Laurent monomials: for exponents other than -1 it follows because a derivative has zero residue; for t^{-1}, the residue of ϕ'/ϕ is 1. Thus the argument is purely formal and does not require analytic convergence.

Apply (5) with $H(x) = e^{yU(x)}$ and set

$$\widetilde{F}_i = [t^i]U(\psi(t)), \qquad 1 \le i \le p.$$

Then $\widetilde{F}_1 = 1$, and the truncation of $U(\psi(t))$ after degree p cannot affect a coefficient of degree p in its exponential. Hence the $\widetilde{F}_i$ satisfy (1).

To check uniqueness directly, suppose $F_2, \dots, F_{i-1}$ are known. The coefficient of y^{p-i+1} on the right of (1) is

$$\frac{F_i}{(p-i)!} + \text{a polynomial in } F_2, \dots, F_{i-1}.$$

Indeed, a contribution involving F_i must consist of that single degree-i factor and $p-i$ degree-one factors. All other contributions use only lower indices. This determines F_i successively for $i = 2, \dots, p$. The leading coefficient in y is already $1/p!$, and the constant coefficient is $a_p = 0$. Therefore $F_i = \widetilde{F}_i$. For $p = 1$ the assertion is just $F_1 = 1$.

The explicit coefficient formula

Formal residue substitution and integration by parts give

$$\begin{aligned}[t^i]U(\psi(t)) &= \mathrm{Res}_x\, U(x)\frac{\phi'(x)}{\phi(x)^{i+1}}\\ &= \frac{1}{i}\,\mathrm{Res}_x\, \frac{U'(x)}{\phi(x)^i} = \frac{1}{i}[x^{i-1}]U'(x)W_p(x)^{i/p}. \end{aligned}$$

The integration-by-parts step uses $\operatorname{Res}_x \frac{d}{dx}(U(x)\phi(x)^{-i}) = 0$. This is the classical Lagrange–Bürmann coefficient identity in the present coordinates. It proves (3) and the conjecture. □

As a check, for $p \geq 2$ the formula gives

$$F_2 = u_2 + \frac{d_1}{p-1}.$$

The displayed constant is the coefficient of $u_1 = 1$; it explains the affine wording when only $u_2, \ldots, u_p$ are treated as free. No constraint or hypothesis from the source has been removed.

Attribution and verification

The setting, unique functions, and conjecture are Federbush's [1]; its Bell-polynomial formulation is credited there to David Williams. Formal residue substitution and Lagrange–Bürmann inversion are classical tools. The present application supplies the coordinate change and the resulting formula for the conjectured functions.

The verifier uses exact rational arithmetic for three parameter profiles at each $1 \leq p \leq 12$. It checks the residue Jacobian, the complete identity as a polynomial in y, a separate triangular reconstruction, compositional inversion, linearity with u_1 fixed, and sensitivity to the condition $a_p = 0$. The finite checks supplement the proof for arbitrary p.

All six pages of the sole current primary version were read. The metadata, bounded public searches, and checked claim indexes did not identify a journal final or a prior full target claim as of September 18, 2026. Review is by the authoring agent only; no independent peer review or proof-assistant certification is claimed.

Tasks 2–5. Proposed follow-up directions

Task 2. Classify when the coefficient transformation in (3) preserves positivity, integrality, or a natural combinatorial interpretation.

Task 3. Describe composition and inversion laws for successive weight changes $A(x)$, including the role of the coefficient constraint.

Task 4. Extend the construction to several variables or other coefficient functionals, and identify the obstructions to an analogous linear transformation.

Task 5. Study growing-index regimes with i/p bounded away from zero, where fixed-index expansions need not be uniform.

References

[1] P. Federbush, *The Genius Conjectures (via Bell Polynomials)*, arXiv:2002.03814v1, 2020. See Sections 1–3, especially Conjecture 1 on page 2.